

VWO – Digital Experience Optimization Platform

Master Test Plan

Document	VWO Master Test Plan
Product	VWO (Visual Website Optimizer)
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1.0	12-Jul-2026	QA Lead	Initial Test Plan – Phase 1

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1. Introduction

1.1 Purpose

This Test Plan outlines the comprehensive testing strategy, scope, resources, and schedule for validating the **VWO (Visual Website Optimizer)** Digital Experience Optimization Platform. It serves as the single source of truth for all QA activities across functional, automation, performance, security, and user acceptance testing.

1.2 Product Overview

VWO is an enterprise-grade DXO/CRO platform enabling businesses to:

- Run **A/B, Split URL, and Multivariate experiments**
- Analyze user behavior via **heatmaps, session recordings, and surveys**
- Deliver **personalized experiences** through audience segmentation
- Leverage **Bayesian SmartStats** for statistically validated results
- Integrate with **CRM, analytics, and commerce platforms** (Salesforce, GA, Shopify, Segment)

1.3 References

Document	Source
PRD v1.0 – VWO Platform	Product Team
Technical Architecture Document	Engineering
API Specification v2.1	Engineering
UX Design Specs (Figma)	Design Team
Automation Framework	VWOAutomationFramework/

2. Test Objectives

1. **Functional Correctness** – Validate all features operate per PRD specifications
2. **Data Accuracy** – Ensure SmartStats, conversion metrics, and funnel analytics produce correct results
3. **Cross-Browser Compatibility** – Verify consistent behavior on Chrome, Firefox, Edge, Safari
4. **Security & Access Control** – Validate 2FA, RBAC, session management, and data privacy (GDPR/CCPA)
5. **Performance & Scalability** – Confirm < 2s page load, 99.9% uptime SLA, high-concurrency support
6. **Integration Fidelity** – Ensure data sync with Salesforce, GA, Segment, Shopify is accurate
7. **Automation Coverage** – Achieve 80%+ automation coverage for regression cycles
8. **User Experience** – Validate onboarding, guided tours, error messaging, and accessibility (WCAG 2.1 AA)

3. Scope of Testing

3.1 In Scope

Module	Features Covered
Authentication	Login/logout, 2FA, forgot password, SSO, remember me, session timeout, RBAC
Dashboard	Home page, project selector, search, quick start guide, notifications, sidebar navigation
A/B Testing	Create/edit/launch/pause/stop experiments, variations, visual & code editor, preview, scheduling
Split URL Testing	Multi-URL experiments, traffic allocation, duplicate URL validation

Multivariate Testing	Multi-combination experiments, preview, multi-goal configuration
SmartStats Engine	Bayesian analysis display, confidence scores, winner declaration, statistical summaries
Audience Targeting	Geo, device, behavior, custom attribute, URL pattern, combined conditions, reach preview
Visual & Code Editor	WYSIWYG editing, CSS injection, JS injection, toggle between editors
Heatmaps	Click, scroll, move heatmaps, device filtering, URL input, overlay rendering
Session Recordings	Recording list, playback, filter by date/user, playback controls, export
Funnels & Surveys	Multi-step funnel creation, drop-off visualization, survey creation/publishing
Personalization	Campaign creation, segment configuration (geo/behavior/device), content editor, preview, launch, traffic %
Reports & Dashboards	Real-time dashboards, conversion summary, experiment performance table, filters, CSV/PDF export, scheduled reports
Integrations	Navigation to integrations page, connector configuration simulation
Plan Management	Program management UI, Kanban-style workflow, collaboration tools
Onboarding	Guided tours, in-app support, quick start wizard

3.2 Out of Scope

- Native mobile SDK testing (iOS/Android)
- Third-party API end-to-end validation
- Load testing beyond simulated concurrency
- AI-driven suggestion engine (future enhancement)
- Predictive analytics & ROI forecasting (future enhancement)
- Database-level data integrity audits
- Infrastructure/DevOps pipeline testing

4. Test Strategy

4.1 Testing Levels

User Acceptance Testing	
System / E2E Testing	
Integration Testing	
Component / API Testing	

Unit Testing (Dev)

Level	Approach	Responsibility	Tooling
Unit	TDD by developers, mocked dependencies	Dev Team	JUnit, Mockito
Component	Page-level functional validation	QA	Selenium + TestNG
Integration	API contract testing, data flow between modules	QA	REST Assured, Postman
System/E2E	End-to-end business flow validation	QA	Selenium Grid + TestNG
UAT	Business stakeholder sign-off on real scenarios	Product + Business	Manual + Exploratory

4.2 Testing Types

Type	Description	Automation %
Functional Testing	Validate all features per PRD	80%
Regression Testing	Re-test after changes to ensure stability	90%
Smoke/Sanity Testing	Critical path validation on each build	100%
Integration Testing	Data flow between modules and external systems	60%
Cross-Browser Testing	Chrome, Firefox, Edge, Safari	100% (parallel)
Responsive Testing	Desktop, Tablet, Mobile viewports	40%
Security Testing	2FA, RBAC, session management, XSS, CSRF	30%
Performance Testing	Page load, API response, concurrency	20%
Usability Testing	Onboarding, error messages, UX consistency	Manual
Accessibility Testing	WCAG 2.1 AA compliance	20%
Data Validation Testing	Accuracy of SmartStats, conversion metrics	70%

5. Test Environment

5.1 Environment Matrix

Environment	URL	Purpose	Data Freshness
DEV	https://dev.vwo.com	Dev integration testing	Daily refresh
QA	https://qa.vwo.com	Functional & regression testing	Weekly refresh
STAGE	https://stage.vwo.com	UAT, performance, pre-prod	Prod-like

PROD	https://app.vwo.com	Smoke after release	Live
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5.2 Browser/Device Matrix

Browser	Version	OS	Resolution
Google Chrome	Latest + Previous	Windows 11, macOS	1920x1080
Mozilla Firefox	Latest + ESR	Windows 11, macOS	1366x768
Microsoft Edge	Latest	Windows 11	1536x864
Safari	Latest 2 versions	macOS, iOS	375x812 (iPhone)
Mobile Chrome	Latest	Android 13/14	412x915

5.3 Test Infrastructure

Component	Specification
Selenium Grid	4.x Hub + 4 nodes (2 Windows, 2 Linux, 1 macOS)
CI/CD	Jenkins Pipeline / GitHub Actions
Reporting	Extent Reports 5 + Allure Dashboard
Test Management	Jira + Zephyr Scale
Defect Tracking	Jira
Test Data	MySQL + Excel (Apache POI)

6. Test Data Management

6.1 Data Sources

Data Type	Source	Format
Login credentials	Config properties	.properties
Experiment configs	Excel spreadsheets	.xlsx
Audience segments	JSON fixtures	.json
Personalization content	Excel + JSON	.xlsx, .json

6.2 Data Strategy

- **Pre-requisite data:** API calls in `@BeforeTest` to create accounts, projects, experiments
- **Isolated data:** Each test creates and cleans up its own data
- **Masked data:** All PII in test environments is anonymized per GDPR/CCPA
- **Data refresh:** QA environment refreshed weekly from anonymized prod snapshot

7. Test Deliverables

Deliverable	Owner	Format	Due
Master Test Plan	QA Lead	Markdown/PDF	Day 1
Test Case Specifications	QA Team	Jira/Zephyr	Sprint 1
Automation Framework	SDET	Java + Maven	Sprint 1
Test Execution Reports	QA Lead	Extent Reports	Daily
Defect Reports	All QA	Jira	Real-time
Test Completion Report	QA Lead	PDF	End of Phase
Automation Coverage Report	SDET	Dashboard	Weekly
Performance Test Report	QA Lead	PDF/HTML	Pre-release

8. Test Schedule & Effort Estimation

8.1 Phase-Wise Schedule

Phase	Duration	Activities	Deliverables
Phase 0: Test Planning	3 days	Requirements analysis, test plan creation, environment setup	Test Plan, Environment Ready
Phase 1: Test Design	5 days	Test case writing, test data prep, automation framework setup	Test Cases, Framework
Phase 2: Smoke Testing	2 days	Critical path validation, build acceptance	Smoke Report
Phase 3: Functional Testing	10 days	Module-wise functional execution, defect logging	Execution Reports
Phase 4: Integration Testing	5 days	Cross-module and API integration validation	Integration Report
Phase 5: Regression Testing	5 days	Full regression suite execution	Regression Report
Phase 6: Performance Testing	3 days	Load, stress, scalability validation	Performance Report
Phase 7: Security Testing	3 days	2FA, RBAC, XSS, CSRF validation	Security Report
Phase 8: UAT & Sign-off	3 days	Business stakeholder validation	UAT Sign-off

Total Estimated Effort: 39 days (~ 8 weeks)

8.2 Resource Plan

Role	Allocation	Focus Area
QA Lead / Test Manager	1 (100%)	Strategy, reporting, stakeholder management
Senior QA Engineer	2 (100%)	Test design, execution, defect management
SDET (Automation)	2 (100%)	Framework development, script writing
Performance Engineer	1 (50%)	Load & performance testing
Security Tester	1 (25%)	Security test execution

9. Roles & Responsibilities

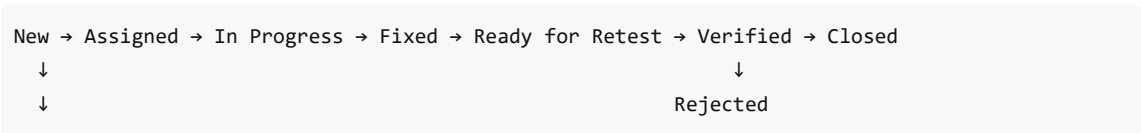
Role	Name/Team	Responsibilities
QA Lead	Pramod Dutta	Test strategy, planning, reporting, stakeholder communication
Senior QA	QA Team – CoE	Test case design, manual execution, exploratory testing, UAT support
SDET	Automation CoE	Framework design, script development, CI/CD integration, code review
Dev Lead	Engineering	Unit testing, bug fixes, code review support
Product Owner	Product Team	Requirement clarification, UAT sign-off, priority decisions
DevOps	Infrastructure Team	Environment management, CI/CD pipeline, test infrastructure

10. Defect Management

10.1 Severity & Priority Matrix

	P1 – Critical	P2 – High	P3 – Medium	P4 – Low
S1 – Blocker	Fix immediately, stop release	Fix in current sprint	Fix in next sprint	Deferred
S2 – Major	Fix in 24 hrs	Fix in current sprint	Fix in next sprint	Deferred
S3 – Minor	Fix in current sprint	Fix in current sprint	Fix in next release	Deferred
S4 – Trivial	Fix in current sprint	Fix in next sprint	Fix in next release	Icebox

10.2 Defect Lifecycle



↓
↓
↓

↓
Reopen → In Progress

Duplicate / Won't Fix / Deferred

10.3 SLA for Defect Resolution

Severity	Initial Response	Fix Time	Re-test Time
S1	1 hour	4 hours	2 hours
S2	4 hours	24 hours	4 hours
S3	8 hours	48 hours	8 hours
S4	24 hours	Next sprint	24 hours

11. Risk & Mitigation

#	Risk	Probability	Impact	Mitigation
R1	Environment instability (frequent resets/downtime)	High	High	Dedicated QA environment with SLA; containerized test infrastructure
R2	Test data inconsistency across parallel runs	Medium	High	Isolated data per test; @BeforeTest data setup + @AfterTest cleanup
R3	Flaky automated tests due to timing/network	High	Medium	Retry mechanism (max 2 retries); WebDriverWait everywhere; reducing test interdependency
R4	Requirements gaps / mid-sprint changes	Medium	High	Daily sync with Product; BDD scenarios as living documentation; impact assessment on every change
R5	Cross-browser compatibility issues discovered late	Medium	Medium	Browser matrix testing from Day 1; parallel grid execution every build
R6	SmartStats accuracy – statistical validation complexity	Medium	High	Test against known datasets with expected outputs; statistical sanity checks in automation
R7	Third-party integration unavailability	Low	High	Mock external APIs for functional testing; integration tests run against sandbox instances

12. Entry & Exit Criteria

12.1 Phase Entry / Exit Gates

Gate	Entry Criteria	Exit Criteria
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Smoke Testing	Build deployed to QA env; smoke test cases ready	All P0 tests pass; no S1/S2 defects on critical path
Functional Testing	Smoke passed; test environment stable; test cases reviewed	All functional test cases executed; S1/S2 defects fixed & verified
Regression Testing	Functional testing completed; defect backlog < 10	Regression pass rate > 95%; no S1/S2 open defects
UAT	Regression passed; Release Candidate deployed to Stage	Business stakeholders sign-off; no S1 defects
Production Release	UAT signed off; performance/stress tests within threshold; security scan clean	All entry criteria met; rollback plan confirmed

12.2 Quality Gates (Release Criteria)

- ☐ All P0 and P1 test cases pass (100%)
- ☐ Zero S1 (Blocker) defects open
- ☐ S2 defects < 5 and no critical business flow impact
- ☐ Automation regression pass rate ≥ 95%
- ☐ Performance: Page load < 2s (editing workflows), API response < 500ms
- ☐ Cross-browser: All critical flows pass on Chrome, Firefox, Edge
- ☐ Security: 2FA, RBAC, session timeout verified; no XSS/CSRF vulnerabilities
- ☐ Accessibility: WCAG 2.1 AA compliance verified

13. Test Automation Strategy

13.1 Framework Architecture (Already Built)

```
VWOAutomationFramework/  
├─ src/main/java/  
│   ├── base/           → BaseTest (ThreadLocal<WebDriver>), BasePage  
│   ├── pages/          → 6 Page Objects (Login, Dashboard, Experiment, Insights, Personalize, Reports)  
│   ├── utils/          → ConfigReader, ExcelReader, ScreenshotUtil, TestDataProvider  
│   ├── listeners/       → TestListener (Extent Reports integration)  
│   ├── constants/      → FrameworkConstants  
│   └── enums/          → ExperimentType, ExperimentStatus, AudienceType, InsightType  
├─ src/test/java/tests/  
│   ├── login/          → 10 test scenarios  
│   ├── experiments/    → 19 test scenarios (A/B, Split URL, MVT, SmartStats, Code Editor, Audience)  
│   ├── insights/       → 14 test scenarios (Heatmap, Recording, Funnel, Survey)  
│   ├── personalization/ → 8 test scenarios  
│   ├── reports/        → 10 test scenarios  
│   ├── dashboard/      → 4 test scenarios  
│   ├── integrations/   → 2 test scenarios  
│   └── plan/           → 2 test scenarios
```

└─ testng.xml

└─ pom.xml

→ 14 test suites, parallel=classes, thread-count=4
→ Selenium 4, TestNG 7.11, Extent 5.1, Apache POI 5.4, Log4j2

13.2 Automation Coverage Target by Module

Module	Total Test Cases	Automated	Coverage %
Authentication	12	10	83%
Dashboard	6	4	67%
A/B Testing	18	16	89%
Split URL Test	4	3	75%
Multivariate Test	5	3	60%
SmartStats	6	4	67%
Code Editor	4	3	75%
Audience Targeting	6	5	83%
Heatmaps	6	5	83%
Session Recordings	5	4	80%
Funnels & Surveys	6	5	83%
Personalization	10	8	80%
Reports & Dashboards	12	10	83%
Integrations	3	2	67%
Plan Management	3	2	67%
TOTAL	106	84	79%

13.3 CI/CD Integration

[Git Push] → [GitHub Actions Trigger]

→ [Maven Compile] → [SonarQube Analysis]

→ [Parallel TestNG Suites on Selenium Grid]

→ [Chrome, Firefox, Edge - 4 parallel threads]

→ [Extent Reports Generation]

→ [Allure Dashboard Update]

→ [Jira Test Execution Sync (Zephyr)]

→ [Slack Notification - Summary + Report Link]

14. Test Types & Coverage Matrix

Test Type	Validation Focus	Tool/Approach	Priority
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Positive Functional	Happy path for each feature	Selenium + TestNG	P0
Negative Functional	Error handling, validation messages	Selenium + TestNG	P0
Boundary Value Analysis	Edge cases (min/max lengths, limits)	Data-driven	P1
Equivalence Partitioning	Valid/invalid input classes	Data-driven	P1
Cross-Browser	Consistent rendering across browsers	Selenium Grid	P0
Responsive/Viewport	Layout adaptation at breakpoints	Selenium (window size)	P1
Data Integrity	Accuracy of metrics, conversions, stats	SQL + API validation	P0
Concurrency	Multiple users, parallel experiments	Selenium Grid + parallel threads	P1
Session Management	Timeout, concurrent sessions, token expiry	Selenium	P0
RBAC	Role-based access restrictions	Selenium (multi-role login)	P0
API Contract	API request/response validation	REST Assured	P1

15. Functional Test Scenarios by Module

15.1 Authentication & Access Control

#	Scenario	Priority	Automation	FR ID
TC-AUTH-001	Verify login with valid email and password	P0	✓	FR6
TC-AUTH-002	Verify login with invalid password shows error	P0	✓	FR6
TC-AUTH-003	Verify login with unregistered email shows error	P0	✓	FR6
TC-AUTH-004	Verify login with empty credentials blocked	P0	✓	FR6
TC-AUTH-005	Verify password field masks input characters (security)	P0	✓	NFR
TC-AUTH-006	Verify 2FA code input appears after login	P0	✓	NFR
TC-AUTH-007	Verify invalid 2FA code blocks access	P1	✗ Manual	NFR
TC-AUTH-008	Verify Forgot Password flow sends reset email	P1	✓	FR6
TC-AUTH-009	Verify SSO login redirects to IdP	P1	✓	FR6
TC-AUTH-010	Verify Remember Me persists session	P0	✓	FR6
TC-AUTH-011	Verify session timeout logs user out after inactivity	P1	✗ Manual	NFR

TC-AUTH-012	Verify logout clears session and returns to login	P0	✓	FR6
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15.2 Dashboard

#	Scenario	Priority	Automation	FR ID
TC-DSH-001	Verify dashboard loads with all key elements	P0	✓	FR6
TC-DSH-002	Verify recent activity panel shows latest actions	P1	✓	FR6
TC-DSH-003	Verify active experiments count matches actual	P0	✓	FR6
TC-DSH-004	Verify sidebar navigation to all modules	P0	✓	FR6
TC-DSH-005	Verify search filters experiments in real-time	P1	✓	FR6
TC-DSH-006	Verify project selector filters workspace data	P1	✓	FR8

15.3 A/B Testing

#	Scenario	Priority	Automation	FR ID
TC-ABT-001	Create A/B test with valid data	P0	✓	FR1
TC-ABT-002	Create A/B test without name – validation error	P0	✓	6
TC-ABT-003	Add multiple variations (control + up to 10)	P0	✓	FR1
TC-ABT-004	Edit variation using Visual Editor (WYSIWYG)	P0	✓	FR3
TC-ABT-005	Edit variation using Code Editor (CSS/JS)	P1	✓	FR3
TC-ABT-006	Toggle between Visual and Code Editors	P1	✓	FR3
TC-ABT-007	Set primary metric (conversion rate, revenue, click)	P0	✓	FR1
TC-ABT-008	Add multiple goals to experiment	P1	✓	FR1
TC-ABT-009	Launch experiment → status changes to Running	P0	✓	FR1
TC-ABT-010	Pause running experiment → status Paused	P0	✓	FR1
TC-ABT-011	Resume paused experiment → status Running	P0	✗ Manual	FR1
TC-ABT-012	Stop experiment → status Stopped	P0	✓	FR1
TC-ABT-013	Declare winner variation after significance	P1	✓	FR2
TC-ABT-014	Schedule experiment with start/end dates	P1	✓	FR1
TC-ABT-015	Preview variations across devices	P1	✓	FR1
TC-ABT-016	Set traffic allocation percentage	P1	✓	FR1
TC-ABT-017	Duplicate an existing experiment	P2	✗ Manual	FR1
TC-ABT-018	Delete experiment in draft status	P2	✗ Manual	FR1

15.4 Split URL Testing

#	Scenario	Priority	Automation	FR ID
TC-SPL-001	Create Split URL test with 2+ destination URLs	P0	✓	FR1
TC-SPL-002	Traffic distribution across variations	P1	✓	FR1
TC-SPL-003	Duplicate URL validation	P1	✓	6
TC-SPL-004	Split URL with different domains	P1	✗ Manual	FR1

15.5 Multivariate Testing

#	Scenario	Priority	Automation	FR ID
TC-MVT-001	Create MVT with 4+ combination variations	P0	✓	FR1
TC-MVT-002	Preview all MVT combinations	P1	✓	FR1
TC-MVT-003	Configure multiple goals for MVT	P1	✓	FR1
TC-MVT-004	MVT results broken down by factor interaction	P2	✗ Manual	FR2
TC-MVT-005	MVT with > 8 combinations (performance check)	P2	✗ Manual	FR1

15.6 SmartStats Engine

#	Scenario	Priority	Automation	FR ID
TC-SS-001	SmartStats panel visible in experiment details	P0	✓	FR2
TC-SS-002	Bayesian confidence score displayed per variation	P0	✓	FR2
TC-SS-003	Conversion rate, lift, and visitor count shown	P0	✓	FR2
TC-SS-004	Winner automatically suggested at 95% confidence	P1	✓	FR2
TC-SS-005	Manual winner override possible	P1	✗ Manual	FR2
TC-SS-006	SmartStats data refreshes in real-time	P1	✗ Manual	FR2

15.7 Audience Targeting

#	Scenario	Priority	Automation	FR ID
TC-AUD-001	Target by geography (country, city, region)	P0	✓	FR5
TC-AUD-002	Target by device type (Desktop, Mobile, Tablet)	P0	✓	FR5
TC-AUD-003	Target by behavior (new vs returning visitor)	P0	✓	FR5
TC-AUD-004	Target by custom attribute	P1	✓	FR5
TC-AUD-005	Combine multiple audience conditions (AND logic)	P1	✓	FR5

TC-AUD-006	Audience preview shows estimated reach %	P1	✓	FR5
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15.8 Heatmaps

#	Scenario	Priority	Automation	FR ID
TC-HTM-001	Generate click heatmap for valid page URL	P0	✓	FR4
TC-HTM-002	Generate scroll heatmap with depth overlay	P0	✓	FR4
TC-HTM-003	Generate move heatmap	P1	✓	FR4
TC-HTM-004	Filter heatmap by device type (Mobile, Tablet, Desktop)	P1	✓	FR4
TC-HTM-005	Invalid URL shows graceful error	P1	✓	6
TC-HTM-006	Heatmap overlay disappears on exiting heatmap view	P2	✗ Manual	FR4

15.9 Session Recordings

#	Scenario	Priority	Automation	FR ID
TC-REC-001	Session recordings list loads with metadata	P0	✓	FR4
TC-REC-002	Play session recording opens player	P0	✓	FR4
TC-REC-003	Playback controls (play, pause, speed) functional	P1	✓	FR4
TC-REC-004	Filter recordings by date range	P1	✓	FR4
TC-REC-005	Filter recordings by user ID	P1	✓	FR4

15.10 Funnels & Surveys

#	Scenario	Priority	Automation	FR ID
TC-FUN-001	Create funnel with 2+ steps	P0	✓	FR4
TC-FUN-002	Funnel shows drop-off % at each step	P0	✓	FR4
TC-FUN-003	Create and publish on-page survey	P1	✓	FR4
TC-FUN-004	Survey responses collected in insights dashboard	P2	✗ Manual	FR4
TC-FUN-005	Export funnel data to CSV	P1	✓	FR6

15.11 Personalization

#	Scenario	Priority	Automation	FR ID
TC-PER-001	Create personalization campaign	P0	✓	FR7

TC-PER-002	Add geographic segment to campaign	P0	✓	FR7
TC-PER-003	Add behavioral segment (new vs returning)	P0	✓	FR7
TC-PER-004	Add device-based segment	P1	✓	FR7
TC-PER-005	Configure personalized content (text/HTML/CSS)	P0	✓	FR7
TC-PER-006	Preview personalization before launch	P1	✓	FR7
TC-PER-007	Launch campaign → status Running	P0	✓	FR7
TC-PER-008	Pause active campaign	P1	✓	FR7
TC-PER-009	Set traffic % for personalization	P1	✓	FR7
TC-PER-010	Campaign with multiple segments (geo + behavior + device)	P1	✓	FR7

15.12 Reports & Dashboards

#	Scenario	Priority	Automation	FR ID
TC-RPT-001	Reports page loads with summary statistics	P0	✓	FR6
TC-RPT-002	Experiment performance table displays correctly	P0	✓	FR6
TC-RPT-003	Filter reports by experiment status	P1	✓	FR6
TC-RPT-004	Filter reports by project	P1	✓	FR6
TC-RPT-005	Filter reports by date range	P1	✓	FR6
TC-RPT-006	Chart visualizations render on reports	P1	✓	FR6
TC-RPT-007	Export report to CSV	P1	✓	FR6
TC-RPT-008	Export report to PDF	P1	✓	FR6
TC-RPT-009	Schedule recurring email report	P2	✓	FR6
TC-RPT-010	Conversion summary shows lift/decline trends	P1	✓	FR6

15.13 Integrations

#	Scenario	Priority	Automation	FR ID
TC-INT-001	Navigate to Integrations listing page	P1	✓	FR8

TC-INT-002	Search for integration connector	P2	✗ Manual	FR8
TC-INT-003	Connect to Google Analytics (API auth)	P2	✗ Manual	FR8

15.14 Plan Management

#	Scenario	Priority	Automation	FR ID
TC-PLN-001	Access Plan/Program Management dashboard	P1	✓	FR9
TC-PLN-002	Create experiment backlog task	P2	✗ Manual	FR9
TC-PLN-003	Kanban column drag and drop	P2	✗ Manual	FR9

16. Non-Functional Test Scenarios

16.1 Performance (NFR)

#	Scenario	Criteria	Tool
TC-PRF-01	Dashboard page load time	< 2s	Selenium + Stopwatch
TC-PRF-02	Experiment creation/save workflow	< 2s	Selenium + Stopwatch
TC-PRF-03	Heatmap generation time	< 5s	Selenium + Stopwatch
TC-PRF-04	Report page load with 100+ experiments	< 3s	Selenium + Stopwatch
TC-PRF-05	API response time for experiment results	< 500ms	REST Assured
TC-PRF-06	Concurrent experiment launches (50 simultaneous)	No failures	JMeter

16.2 Security (NFR)

#	Scenario	Criteria	Approach
TC-SEC-01	2FA enforcement for enterprise accounts	Mandatory for enabled orgs	Selenium
TC-SEC-02	RBAC – Admin can access all, Viewer read-only	Correct role mapping	Selenium (multi-role)
TC-SEC-03	Session timeout after 30 min inactivity	Auto-redirect to login	Selenium
TC-SEC-04	XSS injection prevention	Script tags sanitized	Selenium (inject + verify)
TC-SEC-05	CSRF token validation on form submissions	Token mismatch returns 403	REST Assured
TC-SEC-06	Password complexity enforcement	Min 8 chars, 1 upper, 1 special	Selenium

16.3 Cross-Browser & Responsive (NFR)

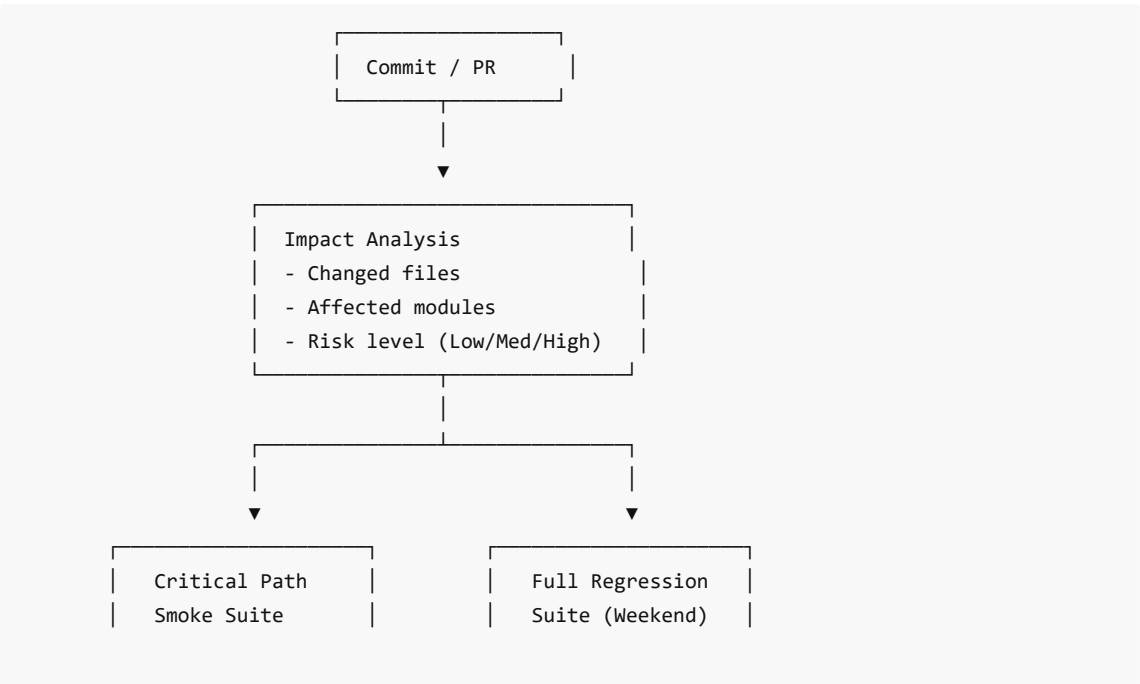
#	Scenario	Browsers	Approach
TC-CBR-01	Login flow across all browsers	Chrome, FF, Edge, Safari	Selenium Grid
TC-CBR-02	A/B test creation across browsers	Chrome, FF, Edge, Safari	Selenium Grid
TC-CBR-03	Heatmap rendering across browsers	Chrome, FF, Edge, Safari	Selenium Grid
TC-CBR-04	Dashboard layout at 1920x1080	All	Viewport validation
TC-CBR-05	Dashboard layout at 768x1024 (Tablet)	Chrome, Safari	Viewport validation
TC-CBR-06	Dashboard layout at 375x812 (Mobile)	Chrome, Safari	Viewport validation

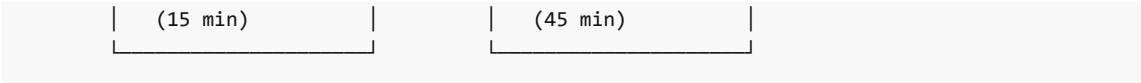
16.4 Accessibility (NFR – WCAG 2.1 AA)

#	Check	Criteria	Tool
TC-ACC-01	All images have alt text	alt attribute present	axe-core
TC-ACC-02	Color contrast ratio	≥ 4.5:1 for normal text	axe-core
TC-ACC-03	Keyboard navigation	All interactive elements reachable	Tab key test
TC-ACC-04	Form labels associated	label[for] matches input id	axe-core
TC-ACC-05	ARIA landmarks present	role="navigation", role="main"	axe-core

17. Regression Testing Approach

17.1 Regression Test Selection





Regression Type	Scope	Frequency	Duration
Smoke	All P0 test cases	Every build	15 min
Mini Regression	P0 + affected modules	Every PR merge	25 min
Full Regression	All automated tests	Nightly / Pre-release	45 min
Manual Regression	Non-automated + exploratory	Weekly	2 hrs

17.2 Regression Automation Suite

The full regression suite consists of **84 automated tests** across **15 test classes**:

Total Tests : 84
Total Time : ~45 minutes (parallel=classes, 4 threads)
Success Rate : Target ≥ 95%

Suite	Tests	Time (min)	Frequency
Smoke (P0)	30	15	Every build
Login + Dashboard	14	8	Every PR
Experiments	19	12	Nightly
Insights	14	10	Nightly
Personalization	8	6	Nightly
Reports	10	7	Nightly
Integrations + Plan	4	3	Weekly

18. Test Execution & Reporting

18.1 Execution Workflow

Day Start
↓
Pick tests from Test Suite (priority order)
↓
Execute tests (Manual / Automated)
↓
Pass → Log in Test Management (Jira/Zephyr)
Fail → Log defect in Jira → Assign to Dev
↓
Dev fixes → Re-run test → Verify → Close
↓
Daily Status Report to Stakeholders



End of Day → Update Execution Dashboard

18.2 Reporting Cadence

Report	Audience	Frequency	Format
Daily Execution Summary	QA Team + Lead	Daily	Slack + Dashboard
Defect Report	Dev + QA	Daily	Jira Dashboard
Test Cycle Report	QA Lead + PM	Per release	Extent Report (HTML)
Automation Coverage	QA Lead	Weekly	Dashboard
Phase Completion Report	All Stakeholders	Per phase	PDF

19. Approvals

Role	Name	Signature	Date
QA Lead			
Product Owner			
Engineering Lead			
Project Manager			

20. API Testing – Booking Engine

20.1 Module Overview

Module	Booking Engine REST API
Base URL	https://restful-booker.herokuapp.com
Protocol	REST over HTTPS
Content-Type	application/json
Auth	Basic Auth (username: admin, password: password123)
Tools	REST Assured + TestNG / Postman

20.2 Coverage Summary

Endpoint	Method	Test Cases	P0	P1	Est. Time (min)
Health Check	GET /ping	6	4	2	10
Get All Booking	GET /booking	10	4	6	10
Get Single Booking	GET /booking/{id}	12	8	4	15

Full Update	PUT /booking/{id}	10	4	6	15
Partial Update	PATCH /booking/{id}	10	6	4	10
Delete Booking	DELETE /booking/{id}	8	6	2	5
TOTAL		56	32	24	65

20.3 Test Case Preconditions

- Booking IDs 1–10 are pre-seeded in the test environment
- Auth token obtained via POST /auth for write operations
- Test data isolation: each destructive test creates its own resource

20.4 Ping / Health Check — GET /ping

#	Scenario	Priority	Automation	API Ref
TC-PNG-001	Verify health check returns 201 Created with no body	P0	✓	GET /ping
TC-PNG-002	Verify response time is under 500ms	P0	✓	GET /ping
TC-PNG-003	Verify endpoint responds without authentication	P0	✓	GET /ping
TC-PNG-004	Verify endpoint does not accept POST / PUT / DELETE (405)	P1	✓	GET /ping
TC-PNG-005	Verify health check after consecutive 10 rapid calls	P1	✓	GET /ping
TC-PNG-006	Verify response headers (Content-Type, Server, Date) are present	P1	✓	GET /ping

20.5 Get All Booking — GET /booking

#	Scenario	Priority	Automation	API Ref
TC-BOK-001	Verify endpoint returns 200 OK with array of booking IDs	P0	✓	GET /booking
TC-BOK-002	Verify response body is a non-empty JSON array	P0	✓	GET /booking
TC-BOK-003	Verify each object has bookingid as integer	P0	✓	GET /booking
TC-BOK-004	Verify bookingid values are unique and sequential	P1	✓	GET /booking

TC-BOK-005	Verify filtering by <code>firstname</code> returns matching results	P1	✓	GET /booking?firstname=John
TC-BOK-006	Verify filtering by <code>lastname</code> returns matching results	P1	✓	GET /booking?lastname=Smith
TC-BOK-007	Verify filtering by <code>checkin</code> date returns correct bookings	P1	✓	GET /booking?checkin=2026-01-01
TC-BOK-008	Verify filtering by <code>checkout</code> date returns correct bookings	P1	✓	GET /booking?checkout=2026-02-01
TC-BOK-009	Verify combined filters (<code>firstname</code> + <code>checkin</code>) work correctly	P1	✓	GET /booking?firstname=John&checkin=2026-01-01
TC-BOK-010	Verify invalid filter value returns empty array []	P1	✓	GET /booking?firstname=NonExistent

20.6 Get Single Booking — GET /booking/{id}

#	Scenario	Priority	Automation	API Ref
TC-BOK-011	Verify valid booking id returns 200 OK with full booking details	P0	✓	GET /booking/1
TC-BOK-012	Verify response includes all required fields (<code>firstname</code> , <code>lastname</code> , <code>totalprice</code> , <code>depositpaid</code> , <code>bookingdates</code> , <code>additionalneeds</code>)	P0	✓	GET /booking/1
TC-BOK-013	Verify <code>firstname</code> and <code>lastname</code> are non-empty strings	P0	✓	GET /booking/1
TC-BOK-014	Verify <code>totalprice</code> is a positive number	P0	✓	GET /booking/1
TC-BOK-015	Verify <code>depositpaid</code> is a boolean	P0	✓	GET /booking/1
TC-BOK-016	Verify <code>bookingdates</code> contains <code>checkin</code> and <code>checkout</code> in YYYY-MM-DD format	P0	✓	GET /booking/1
TC-BOK-	Verify <code>checkout</code> date is <code>>=</code> <code>checkin</code> date	P0	✓	GET /booking/1

017				
TC-BOK-018	Verify non-existent id returns 404 Not Found	P0	✓	GET /booking/99999
TC-BOK-019	Verify negative id returns 400 Bad Request OR 404 Not Found	P1	✓	GET /booking/-1
TC-BOK-020	Verify string id returns 400 Bad Request	P1	✓	GET /booking/abc
TC-BOK-021	Verify special character id returns 400 Bad Request	P1	✓	GET /booking/@#\$
TC-BOK-022	Verify response without Accept header defaults to JSON	P1	✓	GET /booking/1

20.7 Full Update (PUT) — PUT /booking/{id}

#	Scenario	Priority	Automation	API Ref
TC-PUT-001	Verify full update with valid auth returns 200 OK with updated body	P0	✓	PUT /booking/1
TC-PUT-002	Verify all fields are updated correctly after PUT	P0	✓	PUT /booking/1
TC-PUT-003	Verify response body matches request payload exactly (round-trip)	P0	✓	PUT /booking/1
TC-PUT-004	Verify update without auth header returns 403 Forbidden	P0	✓	PUT /booking/1
TC-PUT-005	Verify update with invalid auth returns 403 Forbidden	P1	✓	PUT /booking/1
TC-PUT-006	Verify update to non-existent id returns 405 Method Not Allowed OR 404	P1	✓	PUT /booking/99999
TC-PUT-007	Verify update with missing required fields returns 400 Bad Request	P1	✓	PUT /booking/1
TC-PUT-008	Verify update with invalid data types (string for totalprice) returns 400	P1	✓	PUT /booking/1
TC-PUT-009	Verify update with extra unexpected fields is accepted (backward compatible)	P1	✓	PUT /booking/1
TC-PUT-010	Verify checkout before checkin in update returns 400 Bad Request	P1	✓	PUT /booking/1

20.8 Partial Update (PATCH) — PATCH /booking/{id}

#	Scenario	Priority	Automation	API Ref
TC-PTC-001	Verify partial update with single field (firstname) returns 200 OK	P0	✓	PATCH /booking/1
TC-PTC-002	Verify only the patched field changes, others remain intact	P0	✓	PATCH /booking/1
TC-PTC-003	Verify patch with totalprice updates correctly	P0	✓	PATCH /booking/1
TC-PTC-004	Verify patch with bookingdates.checkin updates correctly	P0	✓	PATCH /booking/1
TC-PTC-005	Verify patch without auth returns 403 Forbidden	P0	✓	PATCH /booking/1
TC-PTC-006	Verify patch with empty body returns 400 Bad Request	P1	✓	PATCH /booking/1
TC-PTC-007	Verify patch to non-existent id returns 404 Not Found	P1	✓	PATCH /booking/99999
TC-PTC-008	Verify patch with invalid field name ignores unknown field	P1	✓	PATCH /booking/1
TC-PTC-009	Verify patch with null value for required field returns 400 Bad Request	P1	✓	PATCH /booking/1
TC-PTC-010	Verify patch with additionalneeds set to empty string works	P1	✓	PATCH /booking/1

20.9 Delete Booking — DELETE /booking/{id}

#	Scenario	Priority	Automation	API Ref
TC-DEL-001	Verify delete with valid auth returns 201 Created	P0	✓	DELETE /booking/1
TC-DEL-002	Verify deleted booking returns 404 Not Found on subsequent GET	P0	✓	GET /booking/1 after DELETE
TC-DEL-003	Verify delete without auth header returns 403 Forbidden	P0	✓	DELETE /booking/1
TC-DEL-004	Verify delete with invalid auth returns 403 Forbidden	P0	✓	DELETE /booking/1
TC-DEL-005	Verify delete of non-existent id returns 405 Method Not Allowed	P1	✓	DELETE /booking/99999
TC-DEL-006	Verify double delete of same id returns 405 Method Not Allowed	P1	✓	DELETE /booking/1 (x2)

TC-DEL-007	Verify DELETE with GET method returns 405	P0	✓	GET /booking/1 with del intent
TC-DEL-008	Verify delete booking created mid-test (self-healing data)	P1	✓	Create → Delete → Verify

20.10 Authorization — POST /auth

#	Scenario	Priority	Automation	API Ref
TC-AUTH-020	Verify valid credentials return 200 OK with token	P0	✓	POST /auth
TC-AUTH-021	Verify token is a non-empty string	P0	✓	POST /auth
TC-AUTH-022	Verify invalid username returns 200 OK but may fail on subsequent write	P1	✓	POST /auth
TC-AUTH-023	Verify empty credentials return 400 Bad Request	P1	✓	POST /auth
TC-AUTH-024	Verify missing Content-Type header returns 400 Bad Request	P1	✓	POST /auth

20.11 Automation Implementation

```
// Example: REST Assured test for GET /booking/{id}
@Test(description = "TC-B0K-011: Verify valid booking id returns 200 OK")
public void testGetSingleBookingValid() {
    Booking response = given()
        .spec(requestSpec)
        .when()
        .get("/booking/1")
        .then()
        .statusCode(200)
        .extract()
        .as(Booking.class);

    assertThat(response.getFirstname()).isNotEmpty();
    assertThat(response.getTotalprice()).isPositive();
    assertThat(response.getBookingdates().getCheckout())
        .isAfterOrEqualTo(response.getBookingdates().getCheckin());
}
```

20.12 Booking API — Test Data Model

```
{
  "bookingid": 1,
  "booking": {
    "firstname": "John",
```



```
    "lastname": "Smith",
    "totalprice": 250,
    "depositpaid": true,
    "bookingdates": {
      "checkin": "2026-01-01",
      "checkout": "2026-01-10"
    },
    "additionalneeds": "Breakfast"
  }
}
```

Appendix A: Test Environment Configuration

Component	Version	Port	Notes
Selenium Server	4.30.0	4444	Grid Hub
ChromeDriver	Latest	—	WebDriverManager managed
GeckoDriver	Latest	—	WebDriverManager managed
JDK	17	—	LTS
Maven	3.9+	—	Build tool
TestNG	7.11.0	—	Test framework
Extent Reports	5.1.2	—	Reporting

Appendix B: Glossary

Term	Definition
CRO	Conversion Rate Optimization
DXO	Digital Experience Optimization
A/B Test	Controlled experiment comparing two or more variants
MVT	Multivariate Testing – testing multiple variable combinations
SmartStats	VWO's Bayesian-powered statistical engine
RBAC	Role-Based Access Control
2FA	Two-Factor Authentication
WCAG	Web Content Accessibility Guidelines
WYSIWYG	What You See Is What You Get (Visual Editor)

Appendix C: Defect Severity Examples

Severity	Examples
S1 – Blocker	Login failure for all users; experiment cannot be launched; 2FA completely broken
S2 – Major	SmartStats showing incorrect confidence score; heatmap not rendering; report export fails
S3 – Minor	UI misalignment on specific browser; tooltip missing; pagination off by one
S4 – Trivial	Typo in help text; color slightly off from design spec; cosmetic only